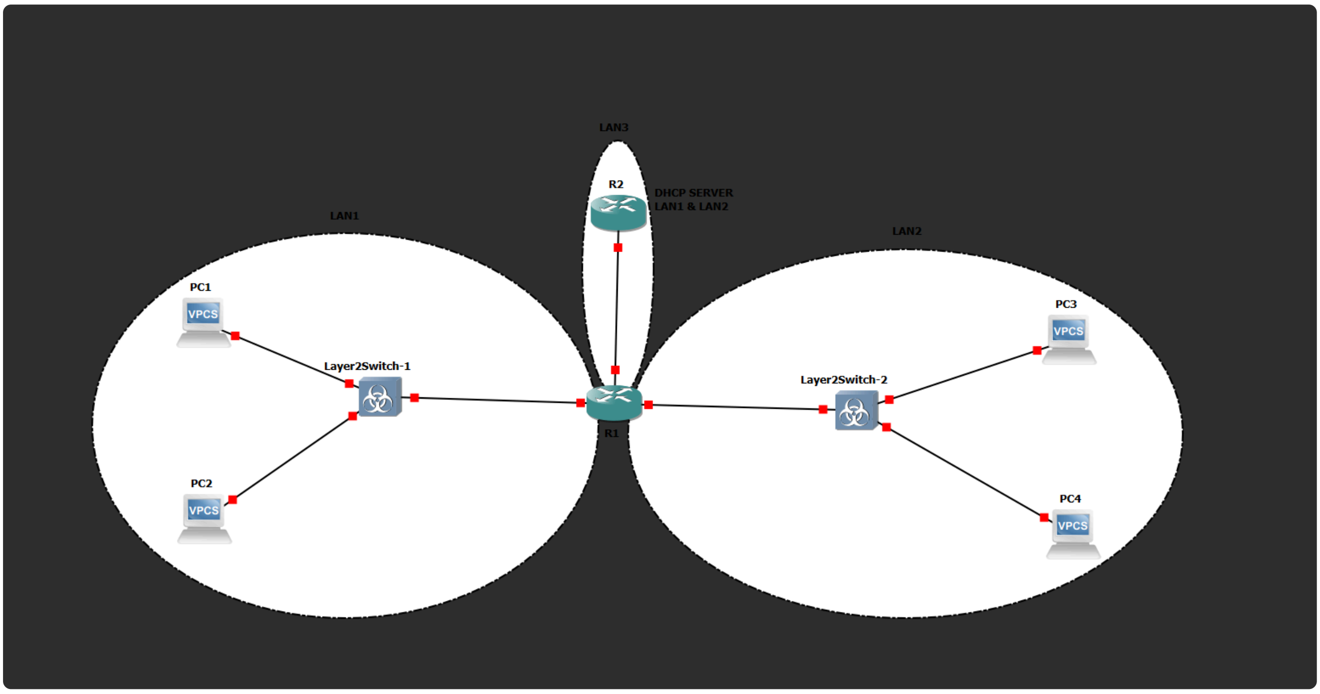
Лабораторная 4

Тема: Настройка протокола DHCP

Все команды для настройки включаются в отчет в текстовом виде, не скриншоты.

nb! - отметка в тексте, "обратите особое внимание"



1) Для заданной на схеме schema-lab4 сети, состоящей из управляемых коммутаторов, маршрутизаторов и персональных компьютеров выполнить планирование и документирование адресного пространства в подсетях LAN1, LAN2, LAN3 и назначить статические адреса маршрутизаторам и динамическое конфигурирование адресов для VPC

План адресации

Выбрана сеть `10.40.4.0/24`. Она разделена с помощью VLSM на две пользовательские подсети `/29` и отдельную транзитную подсеть `/30` между маршрутизаторами.

Подсеть	Назначение	Адрес сети	Маска	Доступные адреса узлов	Broadcast
LAN1	PC1, PC2 и интерфейс R1	10.40.4.0/29	255.255.255.248	10.40.4.1–10.40.4.6	10.40.4.7
LAN2	PC3, PC4 и интерфейс R1	10.40.4.8/29	255.255.255.248	10.40.4.9–10.40.4.14	10.40.4.15
LAN3	Транзитная сеть R1–R2	10.40.4.16/30	255.255.255.252	10.40.4.17–10.40.4.18	10.40.4.19

Назначение адресов устройствам:

Устройство	Интерфейс	Подсеть	Способ настройки	IP-адрес
R1	`FastEthernet0/0`	LAN1	статический	`10.40.4.1/29`
R1	`FastEthernet2/0`	LAN2	статический	`10.40.4.9/29`
R1	`FastEthernet1/0`	LAN3	статический	`10.40.4.17/30`
R2	`FastEthernet0/0`	LAN3	статический	`10.40.4.18/30`
PC1, PC2	`Ethernet0`	LAN1	DHCP	из пула LAN1
PC3, PC4	`Ethernet0`	LAN2	DHCP	из пула LAN2

Настройка маршрутизатора R1:

```
R1#show ip interface brief
*Mar  1 00:03:17.831: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip interface brief
Interface                  IP-Address      OK? Method Status
Prot                ocol
FastEthernet0/0            unassigned      YES unset  administratively down
down
FastEthernet1/0            unassigned      YES unset  administratively down
down
FastEthernet2/0            unassigned      YES unset  administratively down
down
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface FastE
R1(config)#interface FastEthernet0/0
R1(config-if)#description LAN1
```

```
R1(config-if)#ip address 10.40.4.1 255.255.255.248
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Mar  1 00:06:03.835: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed
state t                o up
*Mar  1 00:06:04.859: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthern               et0/0, changed state to up
R1(config)#interface Fast
R1(config)#interface FastEthernet2/0
R1(config-if)#des
R1(config-if)#description LAN2
R1(config-if)#ip ad
R1(config-if)#ip address 10.40.4.9 255.255.255.248
R1(config-if)#no shut
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Mar  1 00:09:57.639: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed
state t                o up
*Mar  1 00:09:58.639: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthern               et2/0, changed state to up
R1(config)#int
R1(config)#interface Fa
R1(config)#interface FastEthernet1/0
R1(config-if)#des
R1(config-if)#description LAN3_TO_R2
R1(config-if)#ip ad
R1(config-if)#ip address 10.40.4.17 255.255.255.252
R1(config-if)#no s
R1(config-if)#no sh
R1(config-if)#no shutdown
R1(config-if)#
*Mar  1 00:10:54.407: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed
state t                o up
*Mar  1 00:10:55.407: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthern               et1/0, changed state to up
R1(config-if)#end
R1#
*Mar  1 00:10:57.607: %SYS-5-CONFIG_I: Configured from console by console
R1#wirte memory
    ^
% Invalid input detected at '^' marker.

R1#write memory
Building configuration...
[OK]
R1#show ip interface brief
Interface                  IP-Address      OK? Method Status
Prot                 ocol
```

```
FastEthernet0/0            10.40.4.1       YES manual up
up
FastEthernet1/0            10.40.4.17      YES manual up
up
FastEthernet2/0            10.40.4.9       YES manual up
up
R1#show ip route connected
     10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.40.4.0/29 is directly connected, FastEthernet0/0
C       10.40.4.8/29 is directly connected, FastEthernet2/0
C       10.40.4.16/30 is directly connected, FastEthernet1/0
```

Настройка маршрутизатора R2:

```
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#inter
R2(config)#interface Fas
R2(config)#interface FastEthernet0/0
R2(config-if)#des
R2(config-if)#description LAN3_TO_R1
R2(config-if)#ip add
R2(config-if)#ip address 10.40.4.18 255.255.255.252
R2(config-if)#no shu
R2(config-if)#no shutdown
R2(config-if)#
*Mar  1 00:13:29.591: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed
state to up
*Mar  1 00:13:30.591: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/0, changed state to up
R2(config-if)#end
R2#w
*Mar  1 00:13:34.279: %SYS-5-CONFIG_I: Configured from console by console
R2#write me
R2#write memory
Building configuration...
[OK]
R2#show ip interface brief
Interface                  IP-Address      OK? Method Status
Protocol
FastEthernet0/0            10.40.4.18      YES manual up
up
R2#show ip route connected
     10.0.0.0/30 is subnetted, 1 subnets
C       10.40.4.16 is directly connected, FastEthernet0/0
R2#
```

Проверка связи м/у маршрутизаторами:

```
R1#ping 10.40.4.18

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.40.4.18, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 60/62/68 ms
R1#
```

-> VPC пока что не находит dhcp сервер, так как он не настроен, т.е. для реализации динамического конфигурирования необходимо выполнить вначале 2й пункт задания.

```
PC1> ip dhcp
DDD
Can't find dhcp server

PC1> ip dhcp
DDD
Can't find dhcp server
```

2) Настроить сервер DHCP на маршрутизаторе R2 для обслуживания адресных пулов адресного пространства подсетей LAN1 и LAN2

Настройка на R2:

```
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#service dhcp
R2(config)#ip dhcp excluded-address 10.40.4.1
R2(config)#ip dhcp excluded-address 10.40.4.9
R2(config)#ip dhcp pool LAN1
R2(dhcp-config)#network 10.40.4.0 255.255.255.248
R2(dhcp-config)#default-router 10.40.4.1
R2(dhcp-config)#exit
R2(config)#ip dhcp pool LAN2
R2(dhcp-config)#network 10.40.4.8 255.255.255.248
R2(dhcp-config)#default-router 10.40.4.9
R2(dhcp-config)#exit
R2(config)#end
R2#
*Mar  1 00:29:48.683: %SYS-5-CONFIG_I: Configured from console by console
R2#write memory
Building configuration...
[OK]
R2#show ip dhcp pool
```

```
Pool LAN1 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 6
 Leased addresses               : 0
 Pending event                  : none
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased addresses
 10.40.4.1            10.40.4.1        - 10.40.4.6         0

Pool LAN2 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 6
 Leased addresses               : 0
 Pending event                  : none
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased addresses
 10.40.4.9            10.40.4.9        - 10.40.4.14        0
R2#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/              Lease expiration        Type
                    Hardware address/
                    User name
```

Настройка на R1:

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface FastEthernet0/0
R1(config-if)#ip helper-address 10.40.4.18
R1(config-if)#exit
R1(config)#interface FastEthernet2/0
R1(config-if)#ip helper-address 10.40.4.18
R1(config-if)#exit
R1(config)#end
R1#wri
*Mar  1 00:33:15.967: %SYS-5-CONFIG_I: Configured from console by console
R1#write memory
Building configuration...
[OK]
R1#show running-config interface FastEthernet0/0
Building configuration...

Current configuration : 144 bytes
!
interface FastEthernet0/0
```

```
 description LAN1
 ip address 10.40.4.1 255.255.255.248
 ip helper-address 10.40.4.18
 duplex auto
 speed auto
end

R1#show running-config interface FastEthernet2/0
Building configuration...

Current configuration : 144 bytes
!
interface FastEthernet2/0
 description LAN2
 ip address 10.40.4.9 255.255.255.248
 ip helper-address 10.40.4.18
 duplex auto
 speed auto
end
```

Далее на VPC:

```
PC1> ip dhcp
DDORA IP 10.40.4.2/29 GW 10.40.4.1

PC1> show ip

NAME        : PC1[1]
IP/MASK     : 10.40.4.2/29
GATEWAY     : 10.40.4.1
DNS         :
DHCP SERVER : 10.40.4.18
DHCP LEASE  : 86392, 86400/43200/75600
MAC         : 00:50:79:66:68:00
LPORT       : 20242
RHOST:PORT  : 127.0.0.1:20243
MTU         : 1500

PC1> save
Saving startup configuration to startup.vpc
.  done
```

-> PC2

```
PC2> ip dhcp
DDORA IP 10.40.4.3/29 GW 10.40.4.1

PC2> show ip
```

```
NAME        : PC2[1]
IP/MASK     : 10.40.4.3/29
GATEWAY     : 10.40.4.1
DNS         :
DHCP SERVER : 10.40.4.18
DHCP LEASE  : 86372, 86400/43200/75600
MAC         : 00:50:79:66:68:01
LPORT       : 20244
RHOST:PORT  : 127.0.0.1:20245
MTU         : 1500

PC2> save
Saving startup configuration to startup.vpc
.  done
```

-> PC3

```
PC3> ip dhcp
DDORA IP 10.40.4.10/29 GW 10.40.4.9

PC3> show ip

NAME        : PC3[1]
IP/MASK     : 10.40.4.10/29
GATEWAY     : 10.40.4.9
DNS         :
DHCP SERVER : 10.40.4.18
DHCP LEASE  : 86377, 86400/43200/75600
MAC         : 00:50:79:66:68:02
LPORT       : 20246
RHOST:PORT  : 127.0.0.1:20247
MTU         : 1500

PC3> save
Saving startup configuration to startup.vpc
.  done
```

-> PC4

```
PC4> ip dhcp
DDORA IP 10.40.4.11/29 GW 10.40.4.9

PC4> show ip

NAME        : PC4[1]
IP/MASK     : 10.40.4.11/29
GATEWAY     : 10.40.4.9
```

```
DNS         :
DHCP SERVER : 10.40.4.18
DHCP LEASE  : 86379, 86400/43200/75600
MAC         : 00:50:79:66:68:03
LPORT       : 20248
RHOST:PORT  : 127.0.0.1:20249
MTU         : 1500

PC4> save
Saving startup configuration to startup.vpc
.  done
```

ip dhcp binding после назначения адресов

```
R2#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/              Lease expiration        Type
                    Hardware address/
                    User name
10.40.4.2           0100.5079.6668.00       Mar 02 2002 12:33 AM
Automatic
10.40.4.3           0100.5079.6668.01       Mar 02 2002 12:34 AM
Automatic
10.40.4.10          0100.5079.6668.02       Mar 02 2002 12:35 AM
Automatic
10.40.4.11          0100.5079.6668.03       Mar 02 2002 12:35 AM
Automatic
R2#show ip dhcp pool

Pool LAN1 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 6
 Leased addresses               : 2
 Pending event                  : none
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased addresses
 10.40.4.4            10.40.4.1        - 10.40.4.6         2

Pool LAN2 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 6
 Leased addresses               : 2
 Pending event                  : none
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased addresses
 10.40.4.12           10.40.4.9        - 10.40.4.14        2
```

```
R2#
```

3) Настроить статическую (nb!) маршрутизацию между подсетями

Настройка на R2:

```
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ip route 10.40.4.0 255.255.255.248 10.40.4.17
R2(config)#ip route 10.40.4.8 255.255.255.248 10.40.4.17
R2(config)#end
R2#write memory
Building configuration...
[OK]
R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static
route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
S       10.40.4.0/29 [1/0] via 10.40.4.17
S       10.40.4.8/29 [1/0] via 10.40.4.17
C       10.40.4.16/30 is directly connected, FastEthernet0/0
R2#
```

На R1 статические маршруты не требуются, поскольку LAN1, LAN2 и LAN3 подключены к нему непосредственно. На R2 настроены статические маршруты в LAN1 и LAN2 через адрес R1 `10.40.4.17`.

4) Проверить работоспособность протокола DHCP и маршрутизации, выполнив ping между всеми VPC

PC1 > ping PC2 and PC3 and PC4

```
PC1> ping 10.40.4.3

84 bytes from 10.40.4.3 icmp_seq=1 ttl=64 time=7.654 ms
84 bytes from 10.40.4.3 icmp_seq=2 ttl=64 time=3.711 ms
```

```
84 bytes from 10.40.4.3 icmp_seq=3 ttl=64 time=3.209 ms
84 bytes from 10.40.4.3 icmp_seq=4 ttl=64 time=6.900 ms
84 bytes from 10.40.4.3 icmp_seq=5 ttl=64 time=6.308 ms

PC1> ping 10.40.4.10

84 bytes from 10.40.4.10 icmp_seq=1 ttl=63 time=29.659 ms
84 bytes from 10.40.4.10 icmp_seq=2 ttl=63 time=16.131 ms
84 bytes from 10.40.4.10 icmp_seq=3 ttl=63 time=15.289 ms
84 bytes from 10.40.4.10 icmp_seq=4 ttl=63 time=15.480 ms
84 bytes from 10.40.4.10 icmp_seq=5 ttl=63 time=15.356 ms

PC1> ping 10.40.4.11

84 bytes from 10.40.4.11 icmp_seq=1 ttl=63 time=28.646 ms
84 bytes from 10.40.4.11 icmp_seq=2 ttl=63 time=14.969 ms
84 bytes from 10.40.4.11 icmp_seq=3 ttl=63 time=15.361 ms
84 bytes from 10.40.4.11 icmp_seq=4 ttl=63 time=15.479 ms
84 bytes from 10.40.4.11 icmp_seq=5 ttl=63 time=15.914 ms
```

PC2 > ping PC3 and PC4

```
PC2> ping 10.40.4.10

84 bytes from 10.40.4.10 icmp_seq=1 ttl=63 time=19.316 ms
84 bytes from 10.40.4.10 icmp_seq=2 ttl=63 time=15.380 ms
84 bytes from 10.40.4.10 icmp_seq=3 ttl=63 time=25.458 ms
84 bytes from 10.40.4.10 icmp_seq=4 ttl=63 time=14.943 ms
84 bytes from 10.40.4.10 icmp_seq=5 ttl=63 time=15.053 ms

PC2> ping 10.40.4.11

84 bytes from 10.40.4.11 icmp_seq=1 ttl=63 time=19.100 ms
84 bytes from 10.40.4.11 icmp_seq=2 ttl=63 time=15.071 ms
84 bytes from 10.40.4.11 icmp_seq=3 ttl=63 time=15.396 ms
84 bytes from 10.40.4.11 icmp_seq=4 ttl=63 time=15.498 ms
84 bytes from 10.40.4.11 icmp_seq=5 ttl=63 time=15.605 ms
```

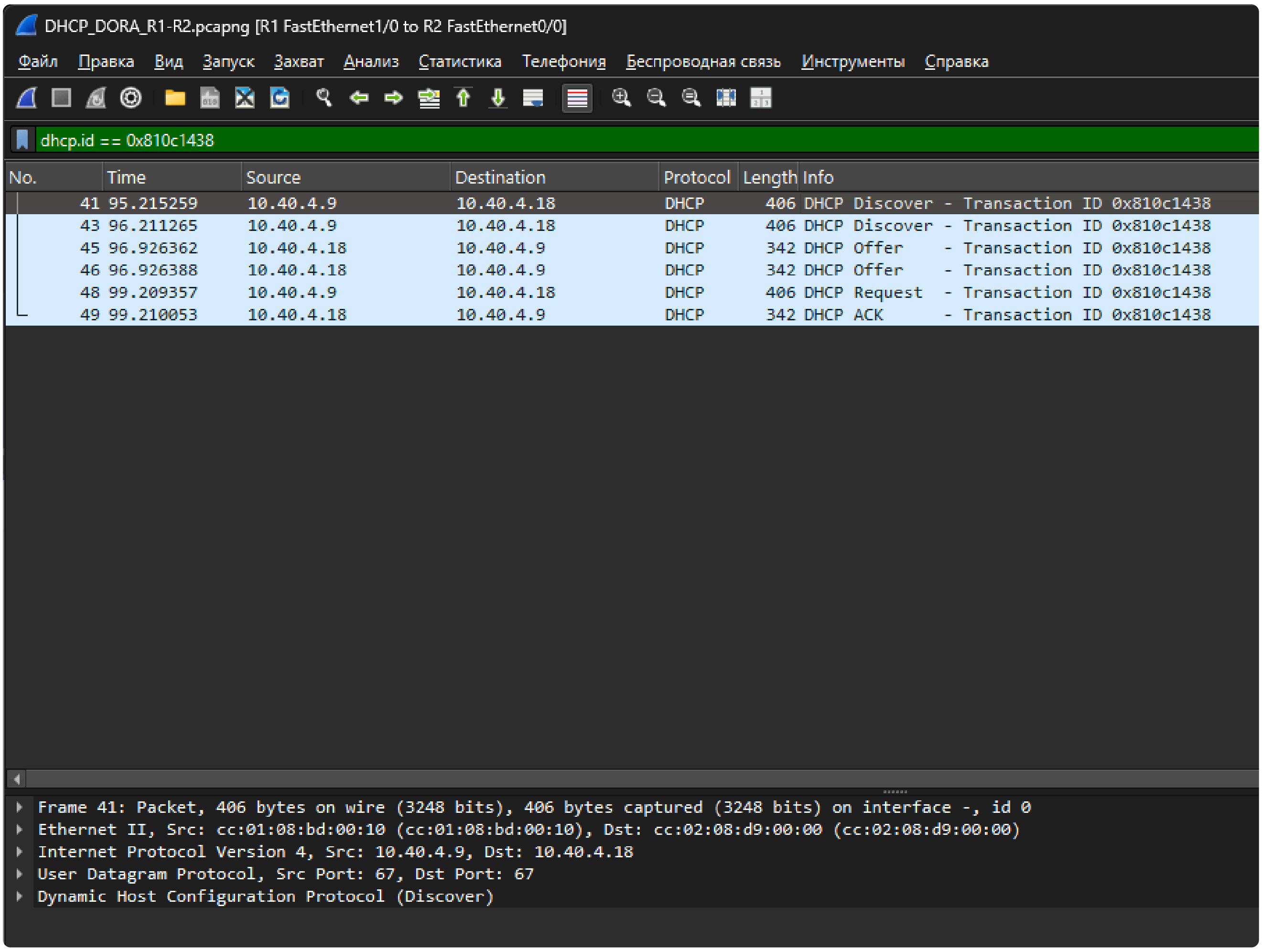
PC3 > ping PC4

```
PC3> ping 10.40.4.11

84 bytes from 10.40.4.11 icmp_seq=1 ttl=64 time=11.830 ms
84 bytes from 10.40.4.11 icmp_seq=2 ttl=64 time=0.805 ms
84 bytes from 10.40.4.11 icmp_seq=3 ttl=64 time=5.442 ms
```

```
84 bytes from 10.40.4.11 icmp_seq=4 ttl=64 time=6.872 ms
84 bytes from 10.40.4.11 icmp_seq=5 ttl=64 time=7.175 ms
```

5) Перехватить в wireshark диалог одного из VPC с сервером DHCP, разобрать с комментариями



Этап	Кадр	Направление	Содержание
Discover	41/43	R1 `10.40.4.9` → R2 `10.40.4.18`	Клиент ищет DHCP-сервер
Offer	45/46	R2 `10.40.4.18` → R1 `10.40.4.9`	Сервер предлагает адрес
Request	48	R1 `10.40.4.9` → R2 `10.40.4.18`	Клиент запрашивает предложенный адрес
ACK	49	R2 `10.40.4.18` → R1 `10.40.4.9`	Сервер подтверждает аренду

```
▸ Frame 41: Packet, 406 bytes on wire (3248 bits), 406 bytes captured (3248 bits) on interface -, id 0
▸ Ethernet II, Src: cc:01:08:bd:00:10 (cc:01:08:bd:00:10), Dst: cc:02:08:d9:00:00 (cc:02:08:d9:00:00)
▸ Internet Protocol Version 4, Src: 10.40.4.9, Dst: 10.40.4.18
▸ User Datagram Protocol, Src Port: 67, Dst Port: 67
▾ Dynamic Host Configuration Protocol (Discover)
    Message type: Boot Request (1)
    Hardware type: Ethernet (0x01)
    Hardware address length: 6
    Hops: 1
    Transaction ID: 0x810c1438
    Seconds elapsed: 0
  ▸ Bootp flags: 0x0000 (Unicast)
    Client IP address: 0.0.0.0
    Your (client) IP address: 0.0.0.0
    Next server IP address: 0.0.0.0
    Relay agent IP address: 10.40.4.9
    Client MAC address: Private_66:68:03 (00:50:79:66:68:03)
    Client hardware address padding: 00000000000000000000
    Server host name not given
    Boot file name not given
    Magic cookie: DHCP
  ▸ Option: (53) DHCP Message Type (Discover)
  ▸ Option: (12) Host Name
  ▸ Option: (61) Client identifier
  ▸ Option: (255) End
    Padding: 00000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
```

Общие поля:

- Transaction ID: 0x810c1438
- Client MAC address: 00:50:79:66:68:03
- Relay agent IP address: 10.40.4.9

  В Discover:
- Option 53: DHCP Discover
- Client IP address: 0.0.0.0
- Your IP address: 0.0.0.0

  В Offer:
- Option 53: DHCP Offer
- Your IP address: 10.40.4.11
- Option 1: Subnet Mask 255.255.255.248
- Option 3: Router 10.40.4.9
- Option 51: IP Address Lease Time 86400 seconds
- Option 54: DHCP Server Indentifier 10.40.4.18

  В Request:
- Option 53: DHCP Request
- Option 50: 10.40.4.11
- Option 54: 10.40.4.18

  В ACK:
- Option 53: DHCP ACK
- Your IP address: 10.40.4.11

- Option 1: 255.255.255.248
- Option 3: 10.40.4.9
- Option 51: 86400 seconds
- Option 54: 10.40.4.18

Захват выполнен на линии R1–R2. Поскольку DHCP-сервер находится в другой подсети, запросы клиента пересылаются маршрутизатором R1 с помощью DHCP Relay. В поле `giaddr` указан адрес интерфейса R1 в LAN2 — `10.40.4.9`, благодаря чему R2 выбирает пул LAN2. Клиент PC4 с MAC-адресом `00:50:79:66:68:03` получает адрес `10.40.4.11/29`, шлюз `10.40.4.9` и время аренды 86400 секунд. Сообщения Discover, Offer, Request и ACK имеют одинаковый Transaction ID `0x810c1438`.

6) Сохранить файлы конфигураций устройств в виде набора файлов с именами, соответствующими именам устройств

Конфигурации всех устройств и файл захвата сохранены

> Полезная информация: возможно, что вам потребуется DHCP Relay